Assignments 3, 2026

Level: BE

Subject : Applied Mathematics

Due Date : Jun 22, 2026

Attempt all questions.

1. Evaluate the integral by using Cauchy Residue Theorem.

   (a) Evaluate the integral $\oint_c \frac{z-23}{z^2-4z-5}\,dz$, where c is a circle $|z-2| = 4$

   (b) Evaluate the integral $\oint_c \frac{z^2 sin\, z}{4z^2-1}\,dz$, where c is a circle $|z| = 2$

   (c) Evaluate the integral $\oint_c \frac{sinh\, z}{2z-i}\,dz$, where c is a circle $|z-i| = 1$

   (d) Evaluate the integral $\oint_c \frac{e^z}{cos\, z}\,dz$, where c is a circle $|z| = 3$

   (e) Evaluate the integral $\oint_c \frac{z cosh\, \pi z}{z^4+13z^2+36}\,dz$, where c is a circle $|z| = \pi$

   (f) Evaluate the integral $\oint_c \frac{e^z}{(z^2+\pi)^2}\,dz$, where c is a circle $|z| = 4$

2. Find the Z-transform of

   (a) $Z(\frac{1}{n!})$, $Z(e_{at})$, $Z(e^{-at}t)$ $Z(\frac{e^{-at}}{n!})$

   (b) Find $Z(e^{-iat})$, $Z(cos\, at)$, $Z(sin\, at)$, $Z(cosh\, at\ sin\, at)$.

3. Evaluate:

   (a) $Z^{-1}\{\frac{z}{(z+1)^2(z-1)}\}$

   (b) Show that $Z^{-1}\{\frac{z^2+z}{(z^2+1)(z-1)} = 1 - cos(n\pi/2)\}$

   (c) $Z^{-1}\{\frac{z^{-1}(1-z^{-2})}{(1+z^{-2})^2}\}$

   (d) $Z^{-1}\{\frac{z(1-e^{-aT})}{(z-1)(z-e^{-at})}\}$

4. Solve the following difference equations:

   (a) $y_{n+2} + 2y_{n+1} + y_n = n$ where $y_0 = y_1 = 0$

   (b) $y_{n+2} - 3y_{n+1} + 2y_n = 4^n$ where $y_0 = 0$ $y_1 = 1$

   (c) $x_{n+1} = 7x_n + 10y_n$, $y_{n+1} = x_n + 4y_n$ where $x_0 = 3$ $y_0 = 2$

   (d) $y_{n+1} - y_{n-1} = u(n)$ for $y_0 = 1$

   (e) $2y_n - 2y_{n-1} + y_{n-2} = u(n)$ where $y_n = 0$ for $n < 0$ and $u(n) = 1$ for $n \geq 0$, $u(n) = 0$ for $u(n) = 0$ for $n < 0$.

   (f) $u_{x+2} - 2cos\,\alpha\ u_{x+1} + u_x = 0$ where $u_0 = 0$ $u_1 = 1$

"BEST Wishes"